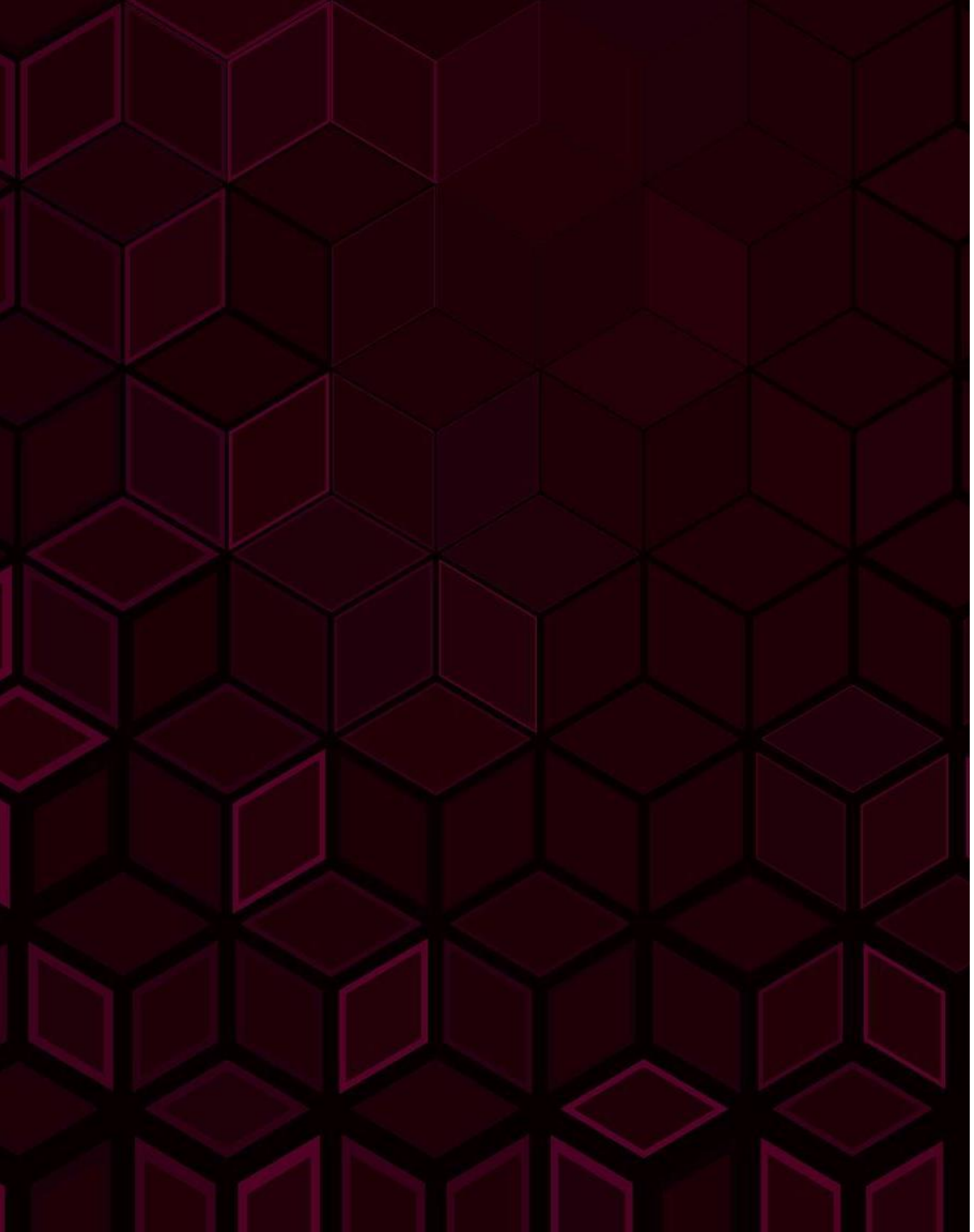


MARS

A MIPS32 Simulator

Introduction

- ***MARS*** (**M**IPS **A**ssembler and **R**untime **S**imulator) is an IDE for MIPS Assembly Language Programming
- MARS is a lightweight interactive development environment (IDE) for programming in MIPS assembly language, intended for educational-level use

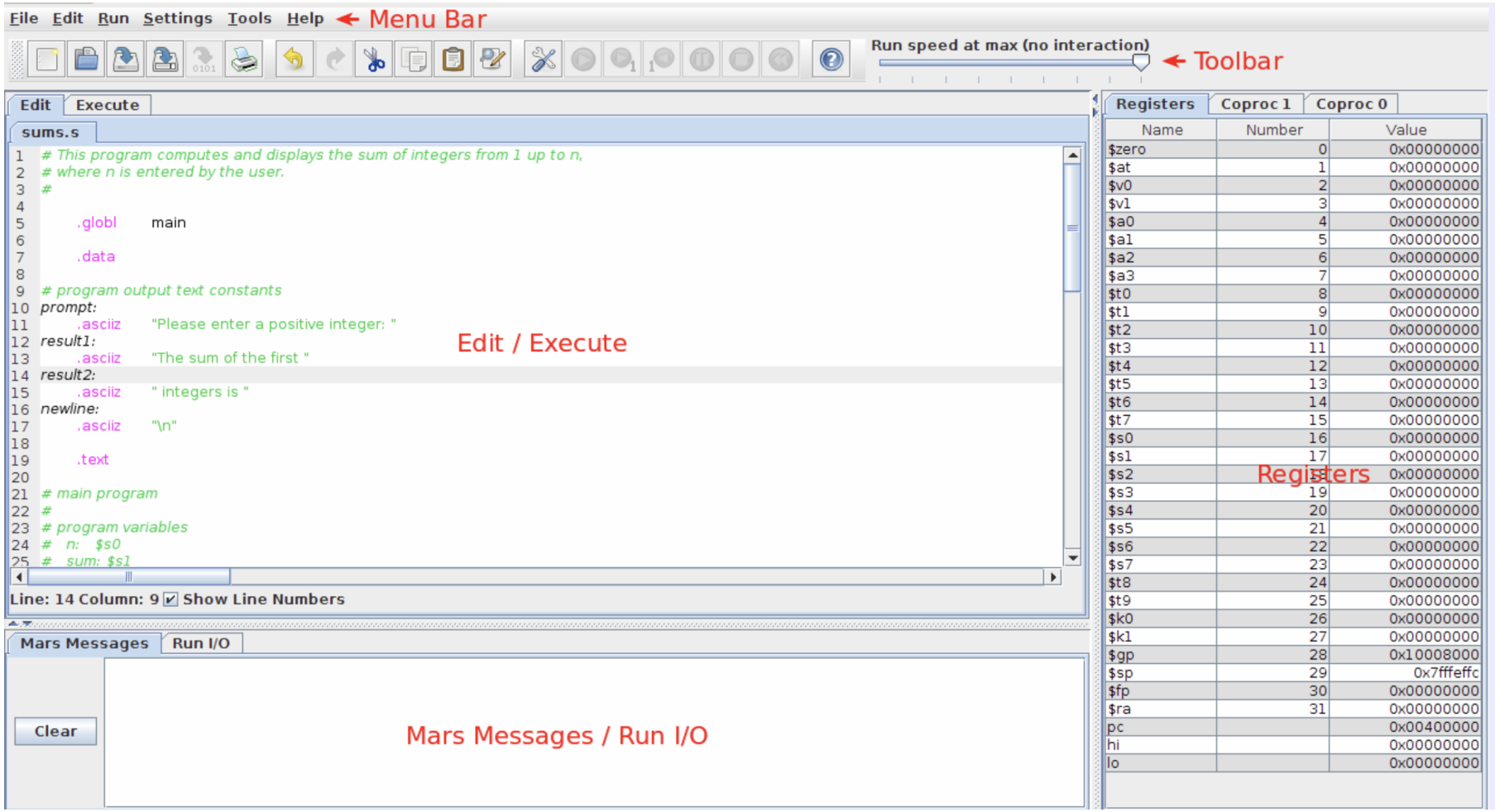
Features

- GUI with point-and-click control and integrated editor
- Easily editable register and memory values, similar to a spreadsheet
- Display values in hexadecimal or decimal
- Variable-speed single-step execution
- Single-step backwards

Installation

- Installation Link - **http://courses.missouristate.edu/KenVollmar/MARS/download.htm**
- Start up the MARS simulator by running/opening (double-clicking) the file you downloaded, Mars_4_2.jar.
- It is a Java executable, so you should have java (i.e. has the Java JRE or JDK) installed on your computer

User Interface



Menu Bar

- The menu bar has menu items for performing various actions.
- The most useful menus for assembly language programming are the "File", "Edit", "Run", and "Help" menus.
- The most commonly used menu items are available on the toolbar.

Tool Bar

- The tool bar contains buttons for the most commonly used menu items.
- They are grouped into four groups corresponding to the "File", "Edit", "Run", and "Help" menus.

File Menu

Menu Item	Icon	Accel	Description
New		Ctrl-N	Create a new empty file
Open …		Ctrl-O	Open a file
Close		Ctrl-W	Close the current file
Close All			Close all files
Save		Ctrl-W	Save the current file
Save as …		Ctrl-S	Save the current file with a new name
Save All			Save all assembly language files
Dump Memory		Ctrl-D	Create a file dump of the text or data section for the current loaded program
Print …			Print the current file
Exit			Exit the Mars program

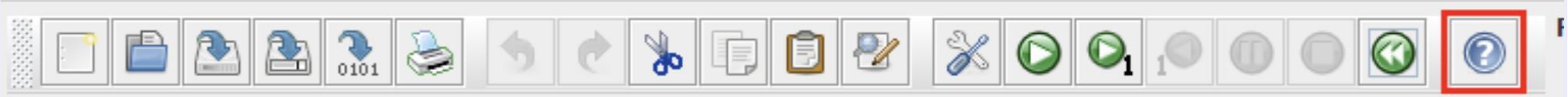
Edit Menu

Menu Item	Icon	Accel	Description
Undo		Ctrl-Z	Undo a previous edit
Redo		Ctrl-Y	Redo a previously undone edit
Cut		Ctrl-X	Cut the currently selected text
Copy		Ctrl-C	Copy the currently selected text
Paste		Ctrl-V	Replace the current text selection with the previous cut or copy text
Find/Replace		Ctrl-F	Find or replace text
Select All		Ctrl-A	Select all text in the current file

Run Menu

Menu Item	Icon	Accel	Description
Assemble		F3	Assemble and load the current file
Go		F5	Run the current loaded program
Step		F7	Run a single instruction of the current loaded program
Backstep		F8	Undo the last executed instruction
Pause		F9	Pause execution of the current loaded program
Stop		F11	Stop execution of the current loaded program
Reset		F12	Reload the current assembled file
Clear All Breakpoints		Ctrl-K	Clear all breakpoints
Toggle All Breakpoints		Ctrl-T	Toggle all breakpoints

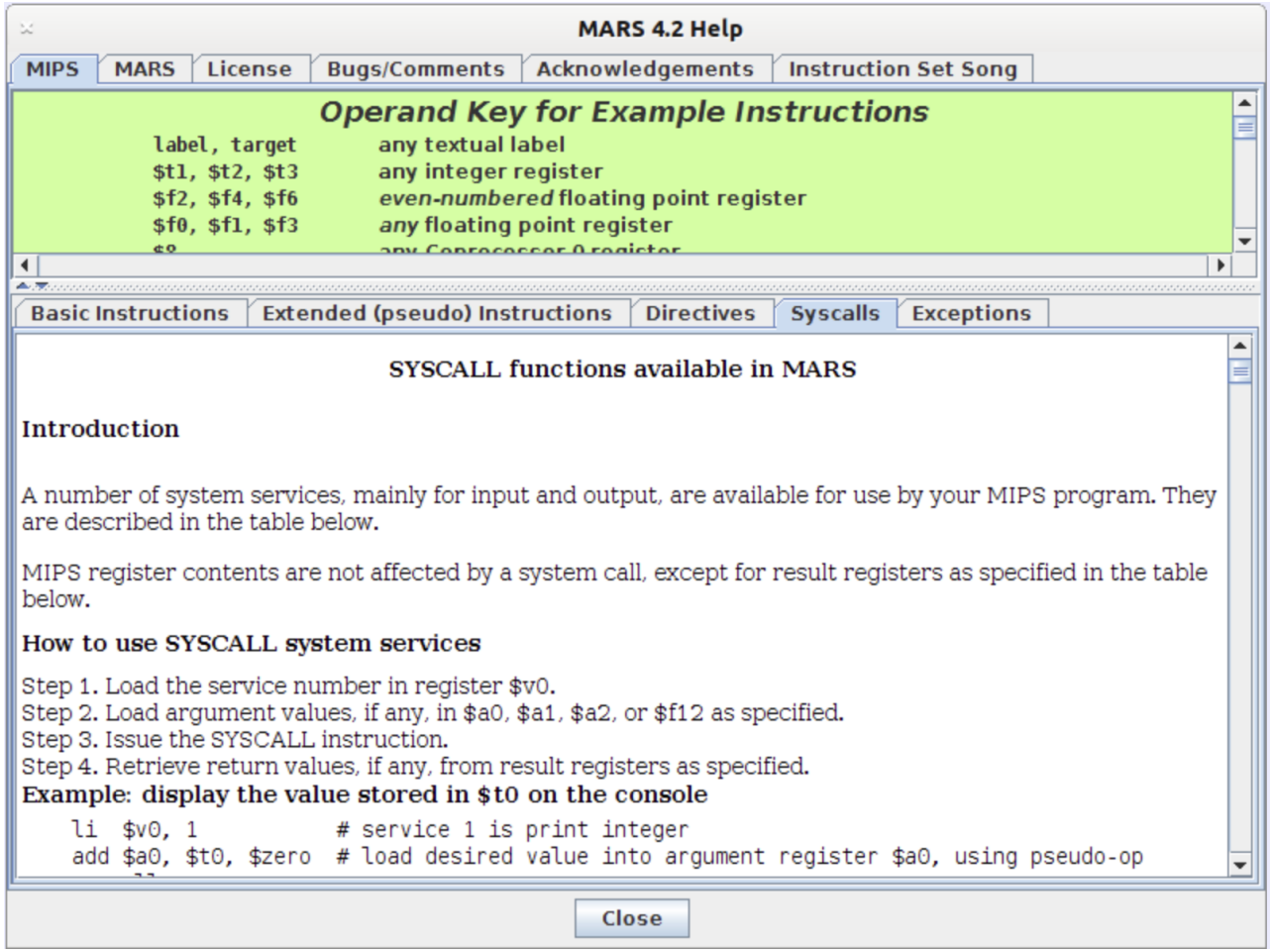
Help Menu



- Much of the information in the help window is available from popups that automatically appear while editing. The major exception is codes for the system calls.

System Calls

This help window screenshot shows the tab selections to bring up the system call information.



Edit/Execute Bar

- The edit/execute pane is a tabbed pane.
- Clicking on the "Edit" tab displays the MIPS assembly language editor.
- Clicking on the "Execute" tab displays assembly and run time information.

Registers Pane

- The registers pane is a tabbed pane. Unless you are writing systems or floating point programs you usually just leave the "Registers" tab selected. This displays the integer registers of the simulated MIPS processor.

Register Naming Conventions

- $zero constant zero
- $at reserved by assembler
- $v0, $v1 for parameter passing
- $a0 to $a3 for arguments
- $t0 to $t9 temporary registers (not saved by callee)
- $s0 to $s7 registers (saved by calle)
- $gp global pointer
- $sp stack pointer
- $ra return address

Mars Messages/Run I/O Pane

The Mars Messages/Run I/O pane is a tabbed pane.

- Clicking on the "Mars Messages" tab displays assembler and some run time messages.
- Clicking on the "Run I/O" tab displays program I/O and some run time messages.

Assembling a File

- Programs can be assembled by clicking "Assemble" Button

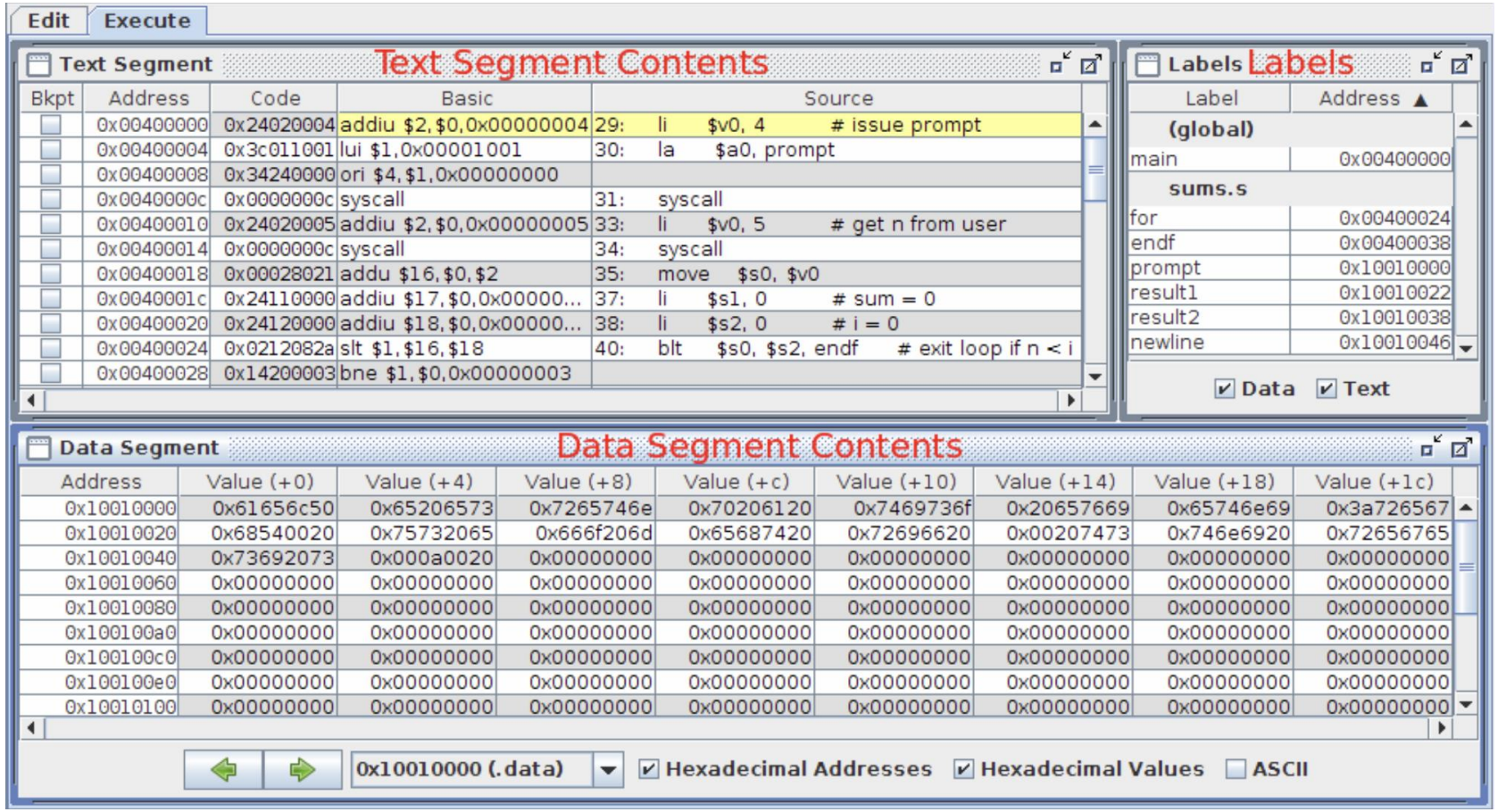
Assembling a File

- Programs can be assembled by clicking "Assemble" Button

- After assembling the program, the Messages/Run I/O pane displays errors if there are any.
- If assembly is successful the Messages/Run I/O pane displays a message indicating that assembly was successful.

Assembling a File (Contd.)

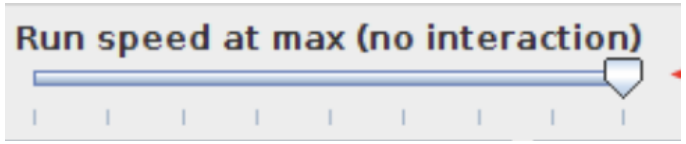
- Execute" tab of the Edit/Execute pane is automatically selected.



Running a Program

- Program is run by clicking on the "Go" go button.

- We can run the program at specified speed using this option – Variable Speed Execution



MIPS32 Assembly Code Layout

```
        .text                   #code section
        .globl main             #starting point, must be global
main:
        #user program code goes here
        .data                   #data section
        #user program data goes here
```

Assembler Directives

a) .text - specifies the user text segment, which contains the instruction.

b) .data - specifies the data segment, where all the data items are defined.

c) .globl sym – specifies that the symbol "sym" is global and can be referred from other files.

d) .word w1, w2,..,wn – Stores the specified 32-bit number in successive memory words.

e) .half h1,h2,…,hn – Stores the specified 16-bit numbers in successive memory half-words.

Assembler Directives

f) .byte b1, b2,..., bn – Stores the specified 8-bit numbers in successive memory bytes.

g) .ascii str – Stores the specified memory(in ASCII Code), but do not null-terminate it.

h) .asciiz str – Stores the specified string in memory(in ASCII code) and null-terminate it

i) .space n – Reserves space for n succivessive bytes in memory.

Example Programs

References

1) http://courses.missouristate.edu/kenvollmar/mars/papers.htm
2) https://www.youtube.com/playlist?list=PL5b07qlmA3P6zUdDf-o97ddfpvPFuNa5A
3) https://www.ece.ucdavis.edu/
4) https://www.d.umn.edu/~gshute/mips/Mars/Mars.html